Sujit Debnath

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SUMMARY

As a CS graduate with experience in teaching, analysis, and volunteering, I enjoy taking the initiative and pursuing new challenges. My research interests focus on integrating AI into various disciplines to solve real-life problems with practical design and implementation.

EDUCATION

Friedrich-Alexander University Erlangen-Nuremberg — **Erlangen, Germany**
Master of Science in Artificial Intelligence — *2021 – Present*

North South University — **Dhaka, Bangladesh**
BS in Computer Science and Engineering; CGPA: 3.66 out of 4.00 (89%-90%) — *2015 – 2018*

- Dissertation Topic: Kothamala - A Bangla Speech Recognition Engine
- Honors in BS: *magna cum laude*, Graduating class of 2019, 22nd Convocation of NSU
- Financial Aid: Awarded a 25% merit-based tuition waiver by NSU

Dhaka College — **Dhaka, Bangladesh**
Higher Secondary Certificate; Science Group; GPA: 5.00 out of 5.00 — *2011 – 2013*

Rampura Ekramunnesa Boys' High School — **Dhaka, Bangladesh**
Secondary School Certificate; Science Group; GPA: 5.00 out of 5.00 — *2009 – 2011*

EXPERIENCE

Working Student - Automation — **Jul 2022 – Present**
Elektrobit Automotive GmbH — *Erlangen, Germany*

- Currently working on the emergency update automation for Volkswagen's electric car.

Project Coordinator — **Feb 2020 – Dec 2020, Apr 2021 – Jan 2022**
***Robi Axiata Limited**, Department of Analytics Center of Excellence* — *Dhaka, Bangladesh*

- Worked on several projects, and some notable projects were technology up-gradation from 3G to 4G (U900 to L900, U21 to L21), L18 integration & LTE bandwidth expansion from 10 to 12 MHz, MIMO integration, and Antenna Optimization.
- Worked on *Microsoft Kaizala* Message Alert project which can catch a few specific degradations of network performance KPI and send a message alert including all the required degradation details to a Kaizala group.
- **Responsibilities:**
  - Analyze network performance KPI of 2G, 3G, and LTE for both Ericsson and Huawei vendors.
  - Develop and tune SQL queries, including SQL based DB reporting.
  - Analyze and performance reporting (e.g., Pre-Post analysis, identifying bad performing cells, worse cells tracking, etc.) of assigned projects.

Lab Officer — **Jan 2019 – Dec 2019**
***North South University**, Department of Electrical & Computer Engineering* — *Dhaka, Bangladesh*

- **Conduct lab classes for the following courses:**
  - CSE115L (Programming Language I Lab) [Fundamentals of Programming Language Using C]
  - CSE215L (Programming Language II Lab) [OOP Using Java]
  - CSE225L (Data Structures and Algorithms Lab) [Using Java]
  - CSE311L (Database Systems Lab) [SQL]
- **Responsibilities:**
  - Designing course curriculum and conducting lab classes.
  - Arrange and evaluate assignments, exams, and projects.
  - Invigilate during exams and assist for grading.

Under-Graduate Assistant — **Jan 2019 – Apr 2019**
***North South University**, Department of Electrical & Computer Engineering* — *Dhaka, Bangladesh*

- **Assist Faculty members for the following course:** CSE115 (Programming Language I), Instructors- *Dr. A. K. M. Ashikur Rahman* & *Dr. Muhammad Asif Hossain Khan*

- **Responsibilities:**
  - Conduct tutorial sessions for students needing extra help outside of class hours.
  - Grading of home-works, assignments and assist faculty members in their course-related work.

PUBLICATIONS

- Shakil Ahmed Sumon, Joydip Chowdhury, **Sujit Debnath**, Nabeel Mohammed, Sifat Momen, **"Bangla Short Speech Commands Recognition Using Convolutional Neural Networks"**, In *1st International Conference on Bangla Speech and Language Processing* (*ICBSLP*), pp. 1-6, 2018.
- Nazmul Alam Diptu, Md. Asif Khan, **Sujit Debnath**, Abdullah Al Imam, Al Mahadi Hasan Rakib, Kazi Asfaq Ahmed Ador, Rashedur M. Rahman, **"Early Detection of Glaucoma Using Fuzzy Logic in Bangladesh Context"**, In *9th International Conference on Intelligent Systems* (*IS'18*), pp. 87-93, 2018.

RESEARCH WORKS

- **Kothamala - A Bangla Speech Recognition Engine:** This research-based project was conducted in two divisions; one was Bengali Short Speech Commands Recognition, and the other one was Bengali Transcription. In the first division, a robust dataset had been collected with 50 classes, and three different Convolutional Neural Network (CNN) architectures had been proposed. In contrast, combined with a language model, another three different model architectures with Connectionist Temporal Classification (CTC) loss function had been experimented for building speech to text engine, where an open-source dataset was taken from *OpenSLR*, released by Google.
- **Early Risk Prediction of Stroke:** In this research-based project, several Machine Learning techniques such as Logistic Regression, Support Vector Classifier, etc along with Synthetic Minority Over-sampling Technique (SMOTE) had been used to predict the risk of stroke on 16 different features excluding the class.

SKILLS

- **Programming Languages:** Python (Frequently used), JAVA, C, C++, SQL
- **Machine Learning, Deep Learning and Visualization Libraries:**
  - Python based libraries: Numpy, Pandas, SciPy
  - Machine Learning library: Scikit-learn
  - Deep Learning library: Keras
  - Visualization: Matplotlib, Plotly, Seaborn
- **Database:** Relational database - Oracle, MySQL, and NoSQL database - MongoDB
- **Version Control System:** GitHub
- **Simulation Tools and Others:** Matlab, Logisim
- **Packages:** MS Word, MS PowerPoint, MS Excel, and LaTeX
- **Languages:** Bengali (Native), and English (Proficient)

EXTRACURRICULAR ACTIVITIES

NSU Problem Solvers (NSUPS) **Dhaka, Bangladesh**
Member *2015 – 2016*

- NSUPS is a problem-solving community based on North South University and this community's primary purpose is to help beginner students get involved in competitive programming. However, I got involved with this community and had the opportunity to participate in several programming competitions organized by NSUPS.

JAAGO Foundation **Dhaka, Bangladesh**
Volunteer *2014 – 2015*

- JAAGO Foundation works for the betterment of the nation through fulfilling the educational needs of underprivileged children. I attended several events organized by JAAGO, including World Environment Day.

Volunteer for Bangladesh (VBD) **Dhaka, Bangladesh**
Community Volunteer *2014 – 2015*

- VBD is a non-political volunteer and youth wing of the JAAGO Foundation, which provides the youth of Bangladesh a voice and platform to contribute to nation-building. However, I attended several events organized by VBD, and some notable events were Universal Children's Day and The Great Kindness Challenge.